6G Cognitive-Hierarchical Swarm Routing (CHSR) Optimization under Dynamic Blockages

Simulation Report

July 19, 2026

1 Introduction

The Terahertz (THz) band (e.g., 300 GHz) provides extreme data rates for 6G networks but suffers from severe propagation losses and susceptibility to physical blockages. This report details the formulations, computational complexity, and performance of a Cognitive-Hierarchical Swarm Routing (CHSR) system driven by Adaptive Particle Swarm Optimization (APSO).

2 System Model and Formulations

2.1 3D Propagation and Attenuation

The spatial domain is discretized into a 2D floor plan of 32×20 meters with a ceiling height of 3.0 m. The Signal-to-Interference-plus-Noise Ratio (SINR) for a user u at coordinates (x_u, y_u, z_u) served by router r at (x_r, y_r, z_r) is determined by the received power P_{rx}.

The base Free-Space Path Loss (FSPL) and molecular absorption are given by:

$$P_{\text{FSPL}} = 20 \log_{10} \left(\frac{4\pi f d_{3D}}{c} \right), \quad P_{\text{abs}} = 4.343 \cdot k(f) \cdot d_{3D} \tag{1}$$

where d_{3D} is the volumetric Euclidean distance, $f = 300$ GHz, and $k(f)$ is the molecular absorption coefficient.

2.2 Static Infrastructure Blockage Model

The environment contains custom static walls (solid concrete, wood, glass). A parametric line equation is drawn from the router to the user. The spatial line intersects horizontal ($y = y_w$) and vertical ($x = x_w$) walls. The static loss L_{static} is the summation of material-specific penalties M_w:

$$L_{\text{static}} = \sum_{w \in W_{\text{intersect}}} M_w \cdot \mathbb{I}(z_i \leq h_w) \tag{2}$$

where z_i is the height of the signal beam at the point of intersection, and h_w is the physical height of the wall.

2.3 Dynamic Human Crowd Density

Human mobility is modeled as a time-varying probability density matrix $D_h(x, y, t)$. The human blockage penalty L_{dynamic} is integrated along the 2D projection of the signal path. If the beam elevation $z(c)$ drops below the human height H_h at grid cell c, the expected penalty is applied:

$$L_{\text{dynamic}} = \sum_{c \in \text{Path}} D_h(c, t) \cdot P_{\text{human}} \cdot \mathbb{I}(z(c) \leq H_h) \tag{3}$$

where $P_{\text{human}} = 20$ dB.

3 Hierarchical APSO Workflow

To minimize mechanical actuation latency, the APSO algorithm operates hierarchically. The fitness function $F(S)$ maximizes coverage proportion Φ, triggering higher tiers only if $\Phi < 0.85$:

- **Tier 1 & 2**: Electronic tuning of beamwidth α and azimuth orientation ϕ.
- **Tier 3**: Vertical translation z adjustments.
- **Tier 4**: Full spatial translation x, y adjustments.

During active optimization, the APSO velocity (v_i) and position (x_i) vector updates for the i-th particle at iteration k are evaluated continuously across the permitted tier dimensions:

$$v_i^{k+1} = w \cdot v_i^k + c_1 r_1 (p_{\text{best},i} - x_i^k) + c_2 r_2 (g_{\text{best}} - x_i^k) \tag{4}$$
$$x_i^{k+1} = x_i^k + v_i^{k+1} \tag{5}$$

where $w = 0.9 - 0.5(k/K_{\max})$ acts as a linearly decaying adaptive inertia weight, c_1, c_2 are cognitive and social coefficients (set to 1.5), and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are random stochastic variables. The vector $p_{\text{best},i}$ represents the particle's historical best state, while g_{best} designates the global optimum discovered by the swarm.

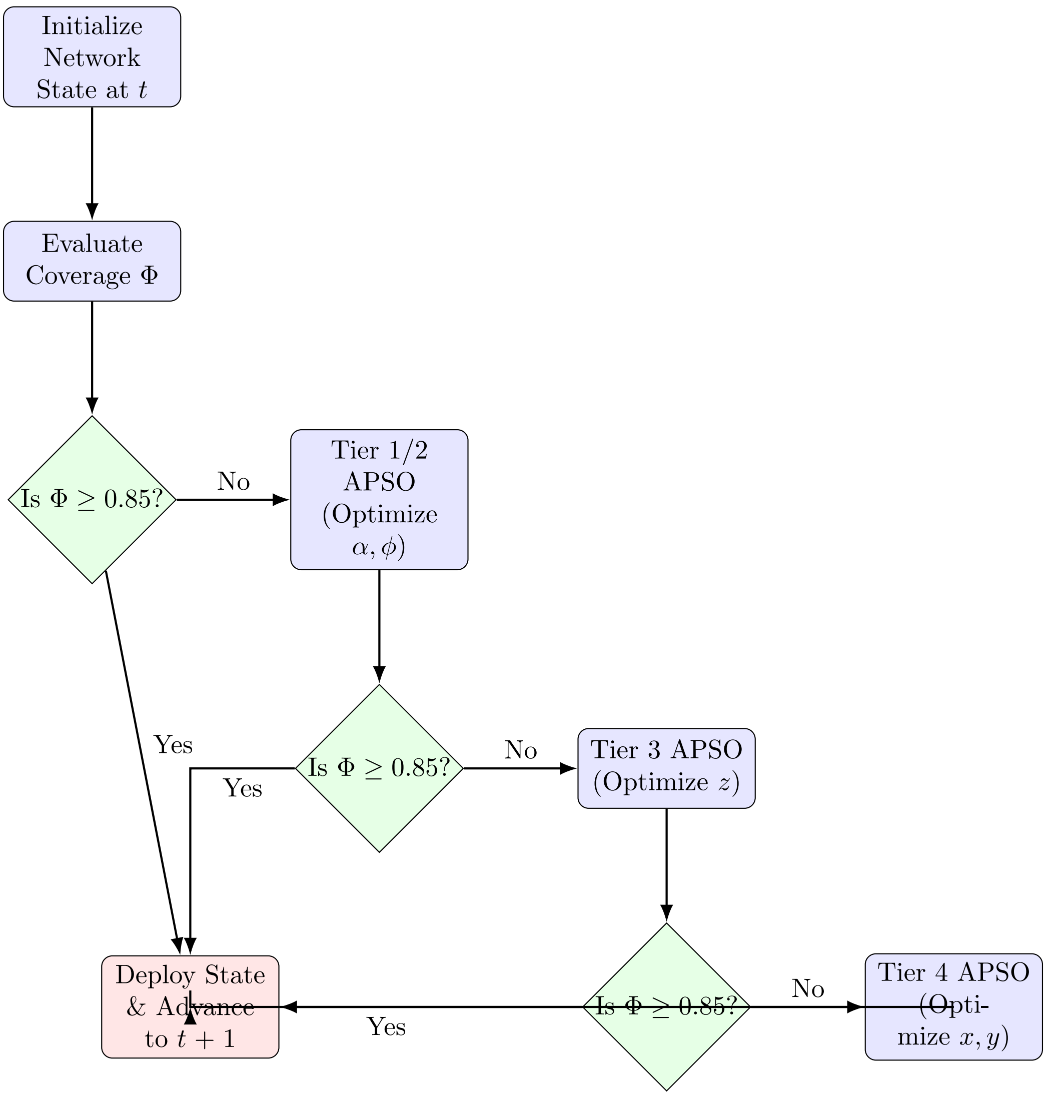

Figure 1: Hierarchical APSO Workflow

4 Time Complexity Analysis

The time complexity of the algorithm is dominated by the ray-tracing required to evaluate the SINR map for the APSO fitness function. Let N be the number of APs, $L \times B$ be the grid dimensions, P be the number of APSO particles, and I be the number of iterations.

- **Ray Tracing & Path Loss Calculation**: For a single router to all grid cells, determining static wall intersections and human density traversals takes $\mathcal{O}(L \cdot B \cdot \max(L, B) \cdot W)$, where W is the number of walls.
- **SINR Evaluation**: Aggregating N routers takes $\mathcal{O}(N \cdot L \cdot B \cdot \max(L, B) \cdot W)$.
- **APSO Execution**: Evaluating P particles over I iterations takes $\mathcal{O}(P \cdot I \cdot N \cdot L^2 \cdot B \cdot W)$ assuming $L \approx B$.

Overall worst-case time complexity per time step (if Tier 4 is reached) is $\mathcal{O}(P \cdot I \cdot N \cdot L^2 \cdot B \cdot W)$. The hierarchical approach drastically reduces the empirical runtime by terminating early at Tier 1/2 whenever possible.

5 Simulation Parameters

Parameter	**Value**
Environment Dimensions	32 m $\times$ 20 m $\times$ 3 m
Operating Frequency (f)	300 GHz
Transmit Power (P_{tx})	20 dBm
Main-Lobe Gain ($G_{\max}$)	30 dBi
Beamwidths (α)	$30°, 45°, 60°, 75°, 90°, 120°$
Candidate Z-axis	2.0 m to 3.0 m (0.2 m step)
Azimuth Increments	45° ($\pi/4$ rad)
Solid/Wood/Glass Penalties	30 dB / 20 dB / 15 dB
System Noise Floor (N_0)	-94 dBm
Human Blocker Penalty	20 dB (Height 1.6 m)

Table 1: System Simulation Parameters

6 Results & Visualizations

The simulation evaluated the custom spatial constraints against shifting human blockages.

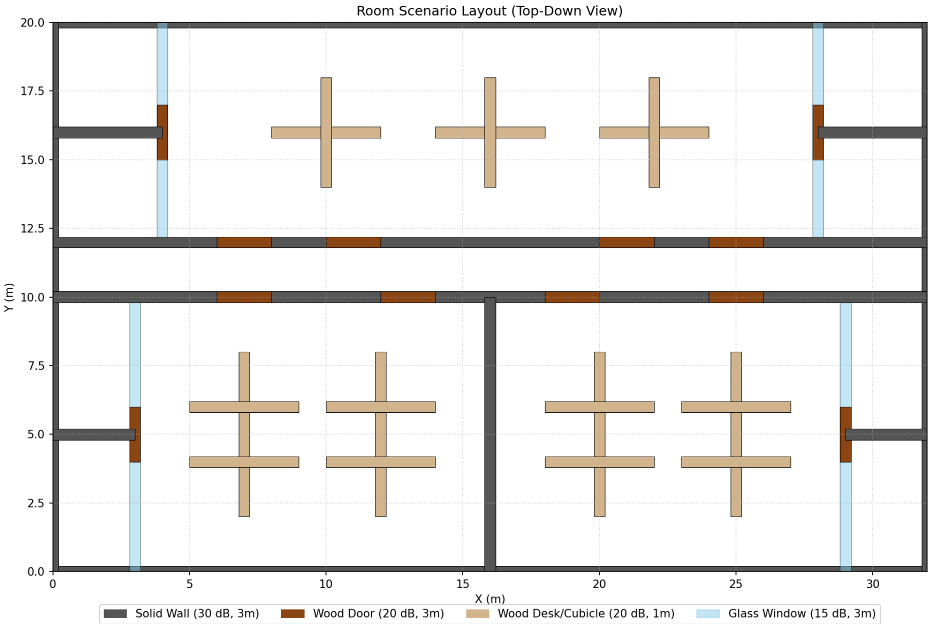

Figure 2: Modeled Room Infrastructure Layout.

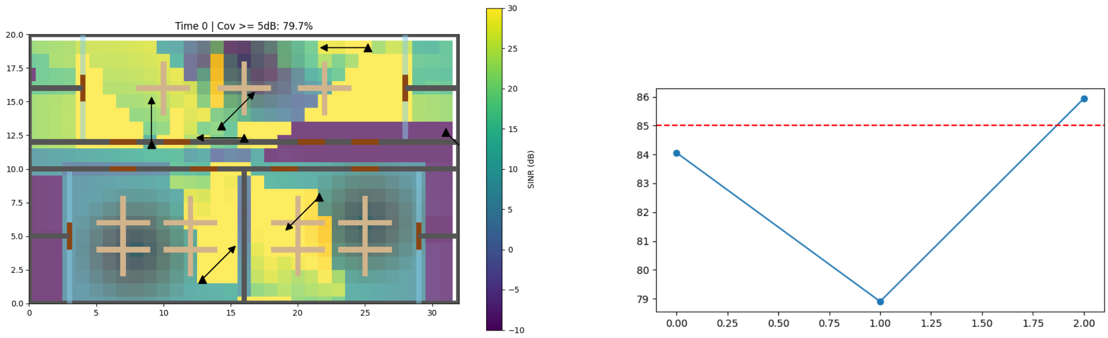

Figure 3: Left: Spatial SINR Heatmap Post-APSO Tier 4 Optimization ($t = 0$). Right: Sustained Coverage Threshold over Time.